

Yicheng Shen

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yshen128@jh.edu | +1 (507) 581-2518 | Website: sheny2.github.io
GitHub: [sheny2](https://github.com/sheny2) | LinkedIn: [yicheng-shen](https://www.linkedin.com/in/yicheng-shen)

Scientific Interests	Real-world evidence synthesis, comparative effectiveness research, and individualized health prediction using observational data, electronic health records, and multi-institutional cohorts.
Methodological Interests	Causal inference, Bayesian hierarchical modeling, multivariate longitudinal modeling, survival analysis, federated and privacy-preserving learning, and network meta-analysis.
Education	Johns Hopkins UniversityBaltimore, MD Ph.D. in Biostatistics, GPA: 4.00/4.00Aug 2024 – May 2029 Advisors: Dr. Jiayi (Jessie) Tong & Dr. Elizabeth A. Stuart Duke UniversityDurham, NC M.S. in Statistical Science, GPA: 3.98/4.00Aug 2022 – May 2024 Advisors: Dr. Hwanhee Hong & Dr. Alexander Volfovsky Carleton CollegeNorthfield, MN B.A. in Statistics & Political Science, GPA: 3.9/4.0Aug 2018 – Jun 2022 Advisors: Dr. Andrew Poppick & Dr. Greg Marfleet Graduate with Distinctions & <i>Magna Cum Laude</i>
Selected Coursework	Statistics: Bayesian statistics, time series analysis, predictive modeling, statistical inference, hierarchical modeling, survival analysis, statistical computing, causal inference, probabilistic machine learning, spatio-temporal modeling, high-dimensional statistics. Mathematics: multivariate calculus, linear algebra, mathematical structure, ordinary differential equations, numerical analysis, computational mathematics, real analysis, measure theoretical probability.
Working Papers & Publications	Shen, Y., Stuart, E. A., Iwashyna, T. J., Hochberg, C., Tong, J. (2026). Scaling Real-World Evidence in Critical Care: A Federated Clone-Censor-Weight Framework for Target Trial Emulation. [Manuscript in preparation for <i>Journal of the Royal Statistical Society: Series C</i>] Shen, Y., Kim, J., Luo, C., Zeger, S. L., Shah, A. A., Tong, J. (2026). MV-PEAL: A Federated Learning Framework for Multivariate Longitudinal EHR Data. [Submitted for the <i>Annals of Applied Statistics</i>] Shen, Y., Kim, J., Luo, C., Zeger, S., Domsic, R. Shah, A., Tong, J. (2026). Unlocking Multi-Institutional Insights into Disease Progression with Federated Learning: PEAL as a Lossless, One-Shot Solution. <i>NPJ Digital Medicine</i>. https://doi.org/10.1038/s41746-026-02861-6

Vazquez, J., **Shen, Y.**, Sanderson, K., Akulian, J., Tong, J., Stuart, E. A.(2026). Federated Learning with Incomplete Data: A Weighted Approach. <https://doi.org/10.48550/arXiv.2605.20125>

Wu, X., **Shen, Y.**, Shan, M., Stuart, E., Lipkovich, I. (2026). Comparing Federated Learning Methods for Estimating Average Treatment Effects in a Target Population from Multiple Real-world Studies and Randomized Trials. [Submitted for *Statistics in Medicine*]

Chao, A., **Shen, Y.**, Ghanta, A., Silver, N., Milardo, R., Santos, C., Tong, J. (2025). Sex Differences in Response to Incretin-Based Medications for Obesity: Systematic Review and Meta-Analysis. [Invited Revision at *Obesity Reviews*]

Shen, Y., Chu, H., Hong, H. (2025). Simulation-Based Methods for Power Calculation in Bayesian Network Meta-Analysis. [Invited revision at *Journal of the Royal Statistical Society Series C*]

Shen, Y., Bail, C., & Volfovsky, A. (2026). Enhancing joint ideal point estimation strategies with Twitter social networks and text data. *Network Science*, 14, e10. [doi:10.1017/nws.2026.10028](https://doi.org/10.1017/nws.2026.10028)

Shen, Y. (2022). Be Steady and Popular: a Modern Counter-Insurgency ABM. *NetLogo Modeling Commons*.

Shen, Y. (2022). Justified Cause? Assessing the Humanitarian Outcomes of US Foreign Aid and Intervention Since the Cold War. *PoliS: Carleton Journal of Political Science*, inaugural edition, pp.129-167.

Shen, Y., Flignor, J., Nachreiner, L., & Wang, K. (2022). Behind the Smoke: An Extreme Value Analysis of Air Pollution in Minnesota. *Consortium for the Advancement of Undergraduate Statistics Education*, spring 2022 issue.

Published Software

Luo, C., Duan, R., ..., **Shen, Y.**, Tong, J., & Chen, Y. (2025). PDA: Privacy-Preserving Distributed Algorithms. R Package. <https://cran.r-project.org/web/packages/pda/index.html>

Shen, Y. (2022). Model of Wheeler. NetLogo Modeling Commons. Northwestern University Center for Connected Learning and Computer-Based Modeling, Evanston. <https://modelingcommons.org/account/models/5618>

Coauthored Chapter

Knoke, D. (2025). Insurgency. In *Network Collective Action: Agent-Based Models of Pandemics, Riots, Social Movements, Insurrections and Insurgencies* (pp. 89-100). Cham: Springer Nature Switzerland.

Honors, Awards & Funding

New England Rare Disease Statistics Workshop Paper Award	2026
ICSA Applied Statistics Symposium Student Paper Award	2026
PHAISE AI Research Day Student Award	2026

NSF Travel Award for the ICSA Applied Statistics Symposium	2026
Honorable Mention for the BEST Award for Master's Research	2024
Teaching Assistant of the Year Award	2024
Dean's Research Award for Master's Students (twice)	2022 & 2023
TADA Travel Award	2023
Duke Career Center Professional Development Fund	2022
First Prize in Undergraduate Statistics Research Project Competition	2022
Phi Beta Kappa Society	2022
Sigma Xi Society	2022
Distinction on Statistics Senior Thesis	2022
Distinction on Political Science Senior Thesis	2022
Dean's Honor List - Top 10% of Class	2021

Teaching Experience

Teaching Assistant (Johns Hopkins University)	Aug 2025 – Present
140.751-2: Advanced Methods in Biostatistics	
140.778: Statistical Computing, Algorithm, and Software Development	
140.779: Advanced Statistical Computing	
140.641: Survival Analysis	
140.622: Statistical Methods in Public Health	
Teaching Assistant (Duke University)	Aug 2023 – May 2024
STA 610: Multilevel and Hierarchical Models	
STA 432: Theory and Methods of Statistical Learning and Inference	
Teaching Assistant (Carleton College)	Sep 2020 – May 2022
STA 320: Time Series Analysis	
STA 230: Applied Regression Analysis	
STA 220: Introduction to Data Science	

Work Experience

Quantitative Resource Center	Northfield, MN
Comprehensive Integrative Exercise Consultant	Sep 2021 – June 2022
Undergraduate Journal of Humanistic Studies	Northfield, MN
Editor of 2020 & 2021 Issues	Mar 2021 – June 2022
Minnesota Council of Nonprofits	Saint Paul, MN
Legislator Data Analytic Internship	Nov 2020 – Dec 2020
Northfield Area Family YMCA	Northfield, MN
Statistical Consultant	Jan 2021 – Mar 2021
Laurence McKinley Gould Library	Northfield, MN
Special Collection and Preservation Assistant	Oct 2018 – Apr 2022

Invited Talks & Conferences

New England Rare Disease Statistics Workshop (NERDS)	2026
Joint Statistical Meeting (JSM)	2026
ICSA Applied Statistics Symposium	2026
Eastern North American Region (ENAR) Spring Meeting	2024, 2026

	Johns Hopkins inHealth Precision Medicine Symposium	2026
	PHAISE AI Research Day	2026
	Duke Industry Statistics Symposium (DISS)	2024, 2026
	Duke Research Data Visualization Competition and Showcase	2024
	Duke Health Data Science Poster Showcase	2023
	New Directions in Analyzing Text as Data (TADA)	2023
	Electronic Undergraduate Statistics Research Conference	2022
	Northfield Undergraduate Mathematics Symposium (NUMS)	2021
	Carleton Undergraduate Research and Internship Symposium	2021
Organized Session	ENAR Invited Session: Federated Learning at the Frontier	2027
Membership	International Chinese Statistical Association New England Statistical Society Eastern North American Region of International Biometric Society International Society for Bayesian Analysis American Statistical Association	
Professional Service & Activity	Reviewer for <i>Epidemiologic Methods</i> Departmental Ph.D. Student Representative Organizer of Biostatistics Education & Student Training Seminar Organizer of Johns Hopkins Causal Inference Working Group Undergraduate Student Departmental Advisor	
Research / Peer Mentoring	Xiao Wu Yifan Zhang Jiawang Wang Dehan Cui	
Skills	Programming Proficiency in: R, Shiny, Python , MySQL, MATLAB, Mathematica, NetLogo . Software & Statistical Implementation Rich experience in GitHub, Quarto, L ^A T _E X & Overleaf, Jupyter Notebook, GWAS with PLINK, coding and operating on server clusters, predictive modeling, hierarchical modeling, agent-based modeling, clustering and classification problems, JAGS, MCMC-related Bayesian sampling and diagnostics.	
Personal Interests	Karate, badminton, movies, hiking, volunteer work, book preservation.	